

NYC YELLOW TAXI

Commercial Performance & Business Analytics

Business Analytics Portfolio Project

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Analysis period	1 Jan 2025 – 31 Jul 2026
Commercial trips	73.38 million
Recorded commercial revenue	~\$2.10 billion
Tools	Python • DuckDB • Power BI (dashboard design)
Data	Real-world NYC TLC trip records

Business question

How can NYC Yellow Taxi operators use recent trip data to understand demand, revenue and operational performance and identify opportunities for improved commercial performance?

1. Executive Summary

The project analyses 19 official NYC TLC Yellow Taxi Parquet files from January 2025 through July 2026. DuckDB was used to query the large dataset efficiently without loading the full raw dataset into Pandas memory.

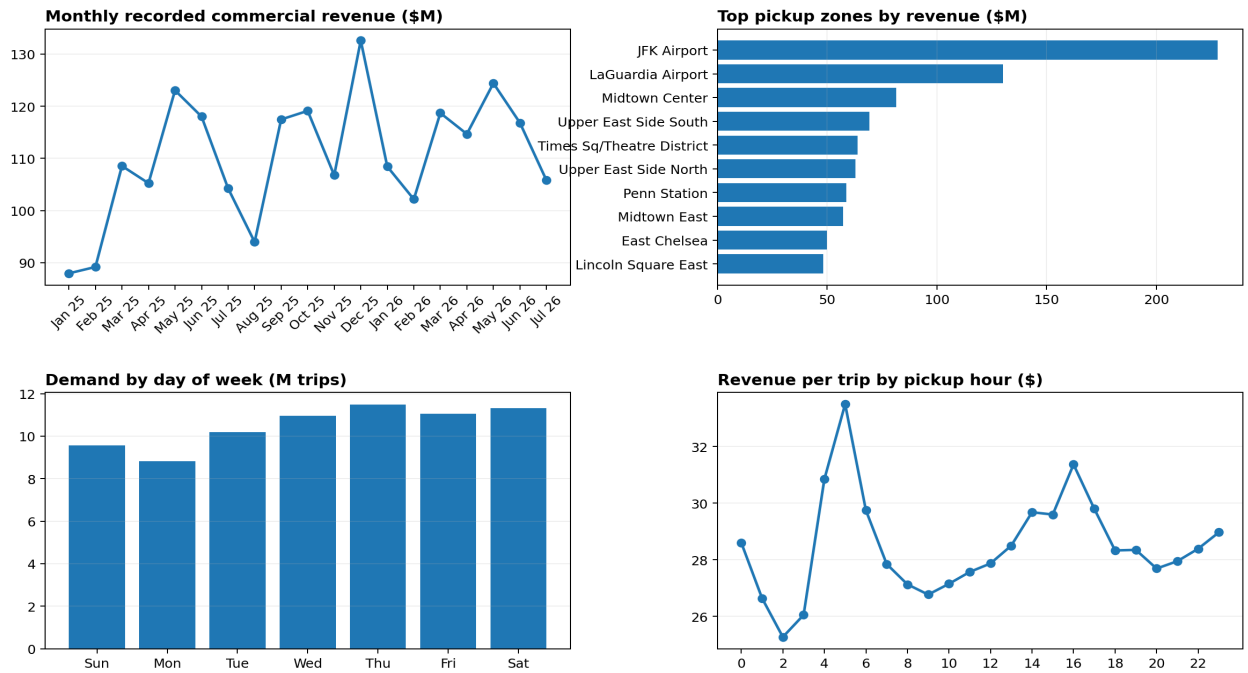
The analysis distinguishes high-volume markets from high-value markets. Manhattan is the dominant volume market, while JFK and LaGuardia are high-value pickup markets.

Short trips can generate stronger recorded revenue relative to trip duration, while longer trips generate more revenue per trip but less revenue per trip-hour.

KPI	Result
Commercial trips	73.38M
Recorded commercial revenue	~\$2.10B
JFK + LaGuardia recorded revenue	\$358.17M
Manhattan revenue	\$1.56B
Queens avg revenue/trip	\$65.23
Peak demand hour	18:00 — 4.72M trips
Peak hourly revenue	17:00 — \$134.76M
<15-min revenue/trip-hour	\$125.50
60+ min revenue/trip-hour	\$52.66
Jan–Jul 2025 avg revenue/trip	\$27.12
Jan–Jul 2026 avg revenue/trip	\$30.35

2. Executive Commercial Overview

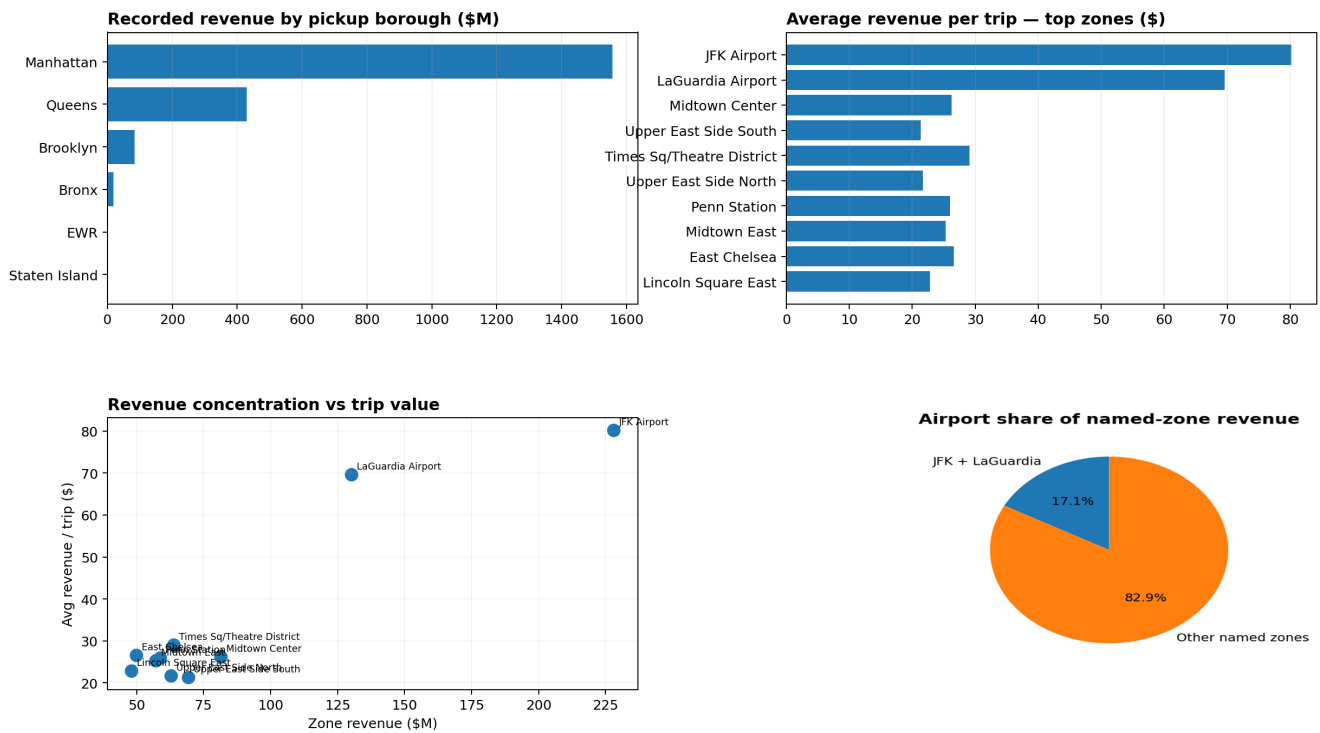
NYC YELLOW TAXI | EXECUTIVE COMMERCIAL OVERVIEW



2. Geographic & Commercial Performance

GEOGRAPHIC & COMMERCIAL PERFORMANCE

GEOGRAPHIC & COMMERCIAL PERFORMANCE



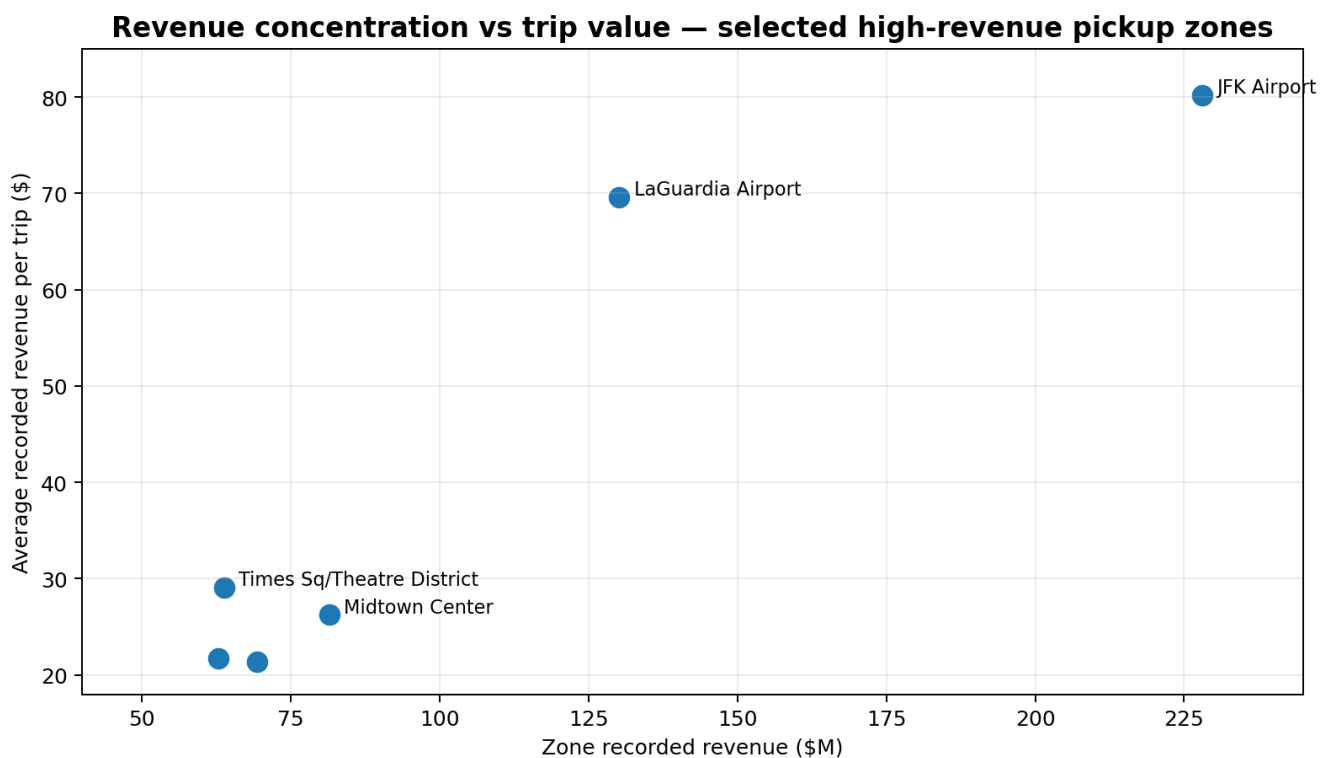
Key commercial interpretation

- Manhattan dominates recorded volume and revenue, while airport markets generate much higher revenue per trip.
- High-volume zones such as Upper East Side South should be assessed using both trip volume and trip value.
- JFK and LaGuardia are distinct high-value pickup markets; their contribution should be monitored separately.
- Revenue shown here is recorded transaction revenue, not profit.

2. Geographic & Commercial Performance

Revenue Concentration vs Trip Value

A zone can be commercially important because of volume, trip value, or both.



HIGH VOLUME + HIGHER VALUE

Midtown Center and Times Sq/Theatre District combine substantial zone revenue with relatively higher recorded revenue per trip.

HIGH VOLUME + LOWER VALUE

Upper East Side South and Upper East Side North generate very large trip volumes, but their average recorded revenue per trip is lower than airport markets.

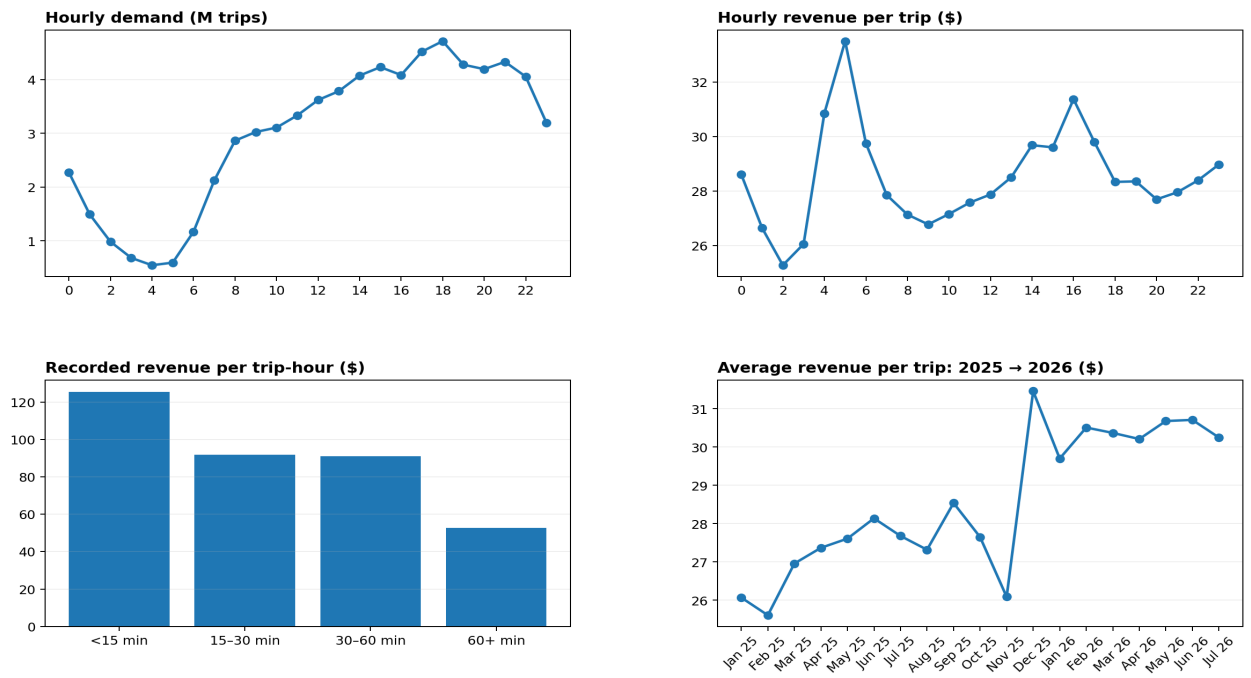
HIGH-VALUE AIRPORT MARKETS

JFK and LaGuardia combine large zone revenue with much higher average recorded revenue per trip.

Business takeaway: Assess zones using both revenue concentration and trip value, not trip volume alone.

2. Demand & Operational Performance

DEMAND TIMING & OPERATIONAL PERFORMANCE



3. Key Analytical Findings

Airports: JFK and LaGuardia generated approximately \$358.17M in recorded commercial revenue. Average recorded revenue was \$80.19 per JFK pickup and \$69.63 per LaGuardia pickup.

Manhattan: 63.33M trips and approximately \$1.56B in recorded revenue make Manhattan the dominant volume market.

Queens: \$429.49M in revenue and \$65.23 average recorded revenue per trip, strongly influenced by airport markets.

Volume vs value: Upper East Side South had 3.24M trips at \$21.37 per trip, while Times Square had 2.19M trips at \$29.09 per trip.

Day of week: Thursday had the highest volume and revenue. Saturday had very high demand but the lowest average revenue per trip at \$26.78.

Operational efficiency: <15-minute trips produced \$125.50 recorded revenue per trip-hour versus \$52.66 for 60+ minute trips.

2026 trip economics: Jan–Jul average recorded revenue/trip increased from \$27.12 to \$30.35 versus Jan–Jul 2025. This should not be interpreted as a causal fare increase without further evidence.

4. Evidence-Based Business Recommendations

Treat airport markets as a distinct commercial segment.

Monitor JFK and LaGuardia separately because their recorded revenue per trip is substantially higher than ordinary Manhattan markets.

Manage Manhattan using both volume and value.

Use both trips and revenue/trip so high-volume, lower-value zones are not treated the same as high-volume, higher-value zones.

Use revenue per trip-hour as an operational KPI.

Assess trip value alongside time utilisation rather than relying only on average revenue per trip.

Plan capacity around peak periods.

The strongest demand occurs in the afternoon/evening, with 18:00 the peak trip-volume hour and 17:00 the peak total-revenue hour.

Monitor the 2026 shift.

Track the higher Jan–Jul 2026 revenue/trip by month, zone and trip type to identify associated commercial conditions.

Limitation: These recommendations are not profitability claims. The dataset does not contain driver costs, fuel costs, waiting time, vehicle availability or complete supply-side information.

5. Methodology & Data Quality

Data: Official NYC TLC Yellow Taxi Trip Records, January 2025 through July 2026.

Processing: Python and DuckDB queried 19 Parquet files; schema differences were handled with union-by-name.

Scope: Pickup timestamps defined the analysis period. Out-of-scope timestamps were excluded.

Commercial analysis: Payment types 3 (No Charge) and 4 (Dispute) were separated from the commercial view because their recorded amounts contain unusual negative transaction patterns.

Distance analysis: Extreme distance records were separated from distance-based operational analysis, while the full commercial dataset was retained for revenue and location analysis.

Location: TLC Taxi Zone Lookup was joined to pickup and drop-off LocationIDs.

Limitations: Passenger counts have substantial missingness; provider-submitted records may contain data-quality issues; recorded revenue is not profit or necessarily cash collected; supply and cost data are unavailable.

Airport-share KPI: JFK + LaGuardia generated \$358.17M, representing approximately 17.12% of named-zone recorded revenue in the full-commercial location view.

6. Portfolio Case Study

Project NYC Yellow Taxi Commercial Performance Analysis

Business problem

Identify where demand and recorded revenue are concentrated, how trip value varies by location and duration, and which patterns can support commercial and operational decisions.

Approach

Real NYC TLC data + DuckDB + Python + Power BI-ready dashboard design.

Analysis covered revenue, demand, geography, airports, time patterns, distance, duration and revenue per trip-hour.

Headline insight

Manhattan provides the largest volume base, airport zones provide high-value trips, and time and location patterns provide actionable commercial segmentation.

Portfolio outputs

- DuckDB analytical workflow for 75M+ real trip records
- Commercial, geographic and operational KPI analysis
- Power BI dashboard visual framework
- Evidence-based business recommendations

Why Yellow Taxi?

Yellow Taxi was selected because it provides a well-defined commercial market with detailed trip-level data covering demand, location, distance, fares, payment types and timing. This makes it suitable for analysing revenue, demand patterns and operational performance within a clearly defined market. The project intentionally focuses on Yellow Taxi rather than combining multiple taxi and for-hire vehicle categories, keeping the analysis commercially focused and interpretable.